Functional Specification & Workflows

Roles, end-to-end processes, business rules, acceptance criteria and user journeys

ConstructionFlow AI | Kosovo-first, multilingual SaaS | Version 1.0

Document purpose

A practical planning reference. All costs, dates, market assumptions, pricing and estimates are illustrative and must be validated with local customers, engineering specialists, legal advisers and actual vendor quotes.

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1. Actors, permissions and operating principles

Actor	Core rights
Organization owner	Subscription, tenant settings, project portfolio
Project manager	Budget, schedules, approval policies, project users
Site engineer	Daily logs, photos, material receipt, issue reporting
Procurement officer	Suppliers, RFQs, quote comparison, purchase orders
Accountant / finance	Invoices, payment status, cost codes, exports
Subcontractor	Restricted assigned tasks and evidence uploads
Auditor / viewer	Read-only scoped reporting and audit trail

- Every record belongs to an organization and optionally a project; access is least-privilege.
- AI suggestions never bypass financial approval or engineer review.

2. End-to-end customer journey

- 1. Owner registers an organization, selects locale, invites employees and creates a project.
- 2. Manager configures project budget, cost codes, milestones and approval thresholds.
- 3. Team imports existing supplier lists, budget spreadsheets and documents.
- 4. Procurement issues RFQs, receives quotations, compares normalized offers and submits a selection.
- 5. Site engineer logs daily progress, materials, blockers and photos, optionally by voice.
- 6. Finance reviews extracted invoices and matches them to orders or deliveries.
- 7. Manager reviews cost and schedule dashboards and approves corrective actions.
- 8. Owner exports a weekly project summary with linked evidence and open risks.

3. Project setup workflow

Trigger and prerequisites

Authorized manager creates a project. Required: project name, location at city level, start date, currency, responsible manager; optional: estimated end date, project category, baseline budget.

Happy path

- Create draft project -> validate dates and permissions -> assign cost-code template -> invite members -> set baseline -> activate.

Exceptions and acceptance criteria

• Duplicate project code is rejected within a tenant; missing required fields show actionable errors.

• Baseline revisions create immutable version history; only authorized roles can approve changes.

• Project summary must show status, latest approved baseline and current commitments.

4. Supplier quotation workflow

Detailed sequence

• Procurement creates RFQ with material description, unit, quantity, delivery date, specification and tax handling.

• Suppliers send PDF/image/Excel quotations; user uploads or forwards into an authorized ingestion channel.

• OCR extracts supplier, item, unit, price, VAT, delivery, currency and payment terms into a review queue.

• Normalize comparable units and flag ambiguous units, missing specifications and inconsistent tax treatment.

• Display side-by-side quotes; compute totals deterministically; highlight cheapest compliant option but never auto-select.

• Procurement submits recommendation -> approver accepts/rejects -> approved PO is generated -> audit log retained.

Exceptions

• Unknown currency exchange rate: require a manually confirmed rate and date.

• Missing unit conversion or concrete grade mismatch: mark quotes non-comparable.

• Duplicate offer, expired validity or supplier withdrawal: warn and prevent silent overwrites.

5. Invoice and material receipt workflow

• 1. Upload invoice and delivery note; virus scan and store original.

• 2. Extract supplier, tax details, invoice number, lines, quantities, unit prices and totals.

• 3. Match supplier and PO; reconcile ordered vs delivered vs invoiced quantities.

• 4. Flag duplicates, mismatches, missing delivery evidence and unusual price differences.

• 5. Finance corrects and confirms; approved record updates actual costs, not merely AI output.

• 6. Export to accounting as CSV/API where available; record sync outcome and retries.

Acceptance criteria

• No invoice can be approved without a traceable source document and reviewer identity.

• Repeated invoice number for the same supplier generates a duplicate warning, with override rationale.

6. Daily site report workflow

• 1. Engineer selects project, site zone, date and shift.

• 2. Engineer types notes or records an Albanian voice memo; attaches photos and material receipts.

• 3. Transcription returns editable text with timestamps where supported.

• 4. AI proposes structured activities, workers, equipment, materials, blockers and safety observations.

• 5. Engineer edits fields and confirms factual accuracy; report is submitted.

• 6. Project manager reviews blockers and creates assigned follow-up tasks.

• 7. Weekly digest aggregates only approved daily reports and links to source entries.

Exceptions

• Low-quality audio or uncertain terms require manual confirmation; never invent quantities.

• Offline submissions should be queued locally only in a future offline-enabled release with encrypted device storage.

7. AI document Q&A workflow

• 1. User uploads approved project documents with classification and access scope.

• 2. Processing extracts text, chunks it, stores embeddings and source offsets.

• 3. User asks a question; search filters documents by tenant, project and permission.

• 4. Assistant produces an answer with citations to document title, page/section and version.

• 5. When evidence is missing or conflicting, it abstains or highlights the discrepancy.

• 6. User can open the exact source, give feedback or flag a wrong answer.

Acceptance criteria

• No answer may cite documents the user is unauthorized to view.

• Answers about contractual responsibility require cited clauses and clear uncertainty language.

8. Dashboards and reporting

Dashboard	Metrics / views
Owner portfolio	Project status, budget baseline, committed, actual, open risks
Project manager	Milestones, delayed tasks, daily logs, pending approvals
Procurement	RFQ cycle time, supplier comparisons, open POs
Finance	Unmatched invoices, expenses by cost code, cash commitments
Site engineer	Today tasks, blockers, drafts, submitted reports

• Budget consumed is not a substitute for physical completion; show them separately and explain data freshness.

• Exports: PDF summary and CSV/XLSX for authorized roles, with redaction where necessary.

9. Functional requirements and user stories

ID	Requirement / acceptance signal
FR-01	Organization and project creation with tenant-safe access
FR-02	Role-based invitations and scoped permissions
FR-03	Budget baseline and revision history
FR-04	RFQ creation, quotation ingestion and normalized comparison
FR-05	Human-approved purchase order creation
FR-06	Invoice extraction and 3-way matching flags
FR-07	Editable Albanian voice transcription to daily report
FR-08	Task creation from approved blockers
FR-09	Permission-filtered RAG with page-level citations
FR-10	Dashboard, filters, export and audit history

• Example user story: As a procurement officer, I can compare three quotations with the same unit and specification so I can recommend a supplier in less time; acceptance: totals are deterministic, missing terms are flagged, and selection requires approval.

10. Non-functional requirements and UX

• Accessibility: keyboard support, sufficient contrast, clear form labels and readable mobile layouts.

• Localization: Albanian first; translations use reviewed construction vocabulary; dates, decimal separators and EUR formatting are consistent.

• Performance target: typical dashboard under 3 seconds at pilot load; async documents show processing status and retry actions.

• Availability objective for pilot: 99.5% monthly excluding announced maintenance; reassess before contracted SLA.

• Security: least privilege, MFA for administrators, audit logs, encrypted transport and storage.

• Retention and deletion: configurable policies, legal-hold exceptions where applicable, and documented export.

• AI quality: measure extraction field precision, transcription correction rate and unsupported answer rate by document type.

11. UAT scripts and release criteria

Test	Expected result
Cross-tenant user opens foreign project	Denied, logged, no metadata leak
Quotation uses inconsistent units	Comparison blocked until normalized
Invoice repeated	Duplicate warning with review workflow
Unclear Albanian audio	Editable transcript with uncertainty
RAG question without evidence	No fabricated clause or page citation
Project manager approves PO	Approval identity and version recorded
Export by restricted subcontractor	Only authorized project scope included

• Pilot exit criteria: 90%+ completion of priority workflows in supervised UAT, zero critical cross-tenant defects, verified backup restoration and signed acceptance from design partners.